LINUX / SYSTEMS

firewalld & nftables Visual Guide

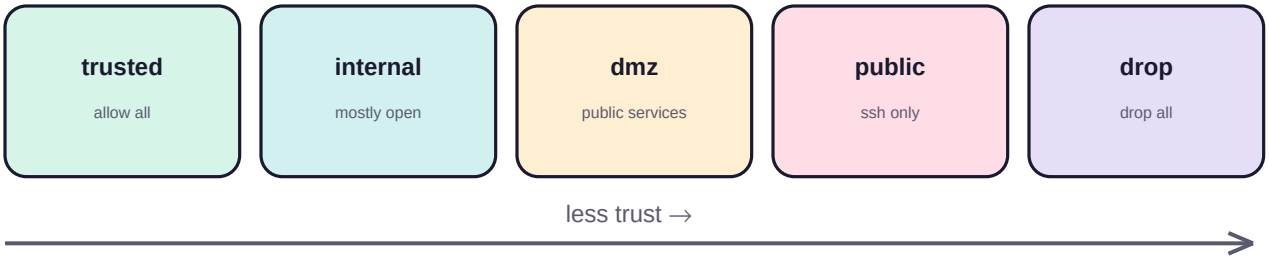
Zones, services vs ports,
rich rules, and nftables chains.

ANIMATED WALKTHROUGH
Step through pages to play the animation

STEP 1/6

Zones: Trust Levels for Interfaces

Each interface lives in ONE zone; the zone defines the default posture.




```
$ sudo firewall-cmd --get-active-zones
$ sudo firewall-cmd --zone=public --add-interface=eth0 --permanent
$ sudo firewall-cmd --get-default-zone
```

A laptop on café Wi-Fi belongs in 'public'; the same laptop on the office LAN in 'work'. Zones make the firewall follow the network, not the machine.

STEP 2/6

Services vs Ports

Services are named port-sets — self-documenting and safer.

service: http

ports 80/tcp · maintained

port: 8080/tcp

raw number · your problem

```
$ sudo firewall-cmd --zone=public --add-service=http --permanent
$ sudo firewall-cmd --zone=public --add-port=8080/tcp --permanent
$ firewall-cmd --get-services | tr ' ' '\n' | grep -i sql
$ sudo firewall-cmd --reload
```

- --add-service=http opens whatever the service definition says — if it changes upstream, you inherit it correctly.
- --add-port is for YOUR custom apps. Name them in comments or a custom service XML.
- List services before inventing ports: most standard things already have a name.

STEP 3/6

Rich Rules: Surgical Exceptions

When a service/port is too blunt: match source, log, rate-limit.

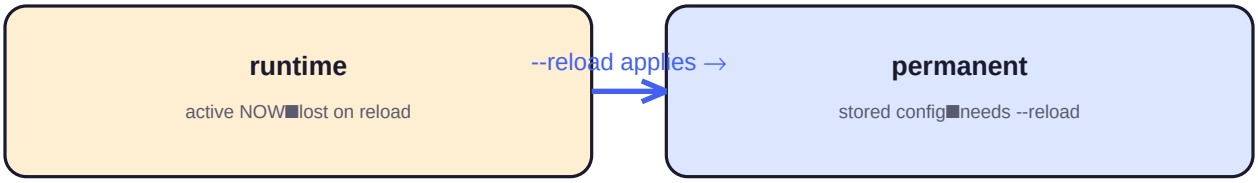
```
$ sudo firewall-cmd --zone=public --permanent \
  --add-rich-rule='rule family=ipv4 source address=10.0.0.0/8 \
  port port=22 protocol=tcp accept'
$ sudo firewall-cmd --permanent \
  --add-rich-rule='rule service name=ssh log prefix="ssh: " \
  limit value="5/m" accept'
```

- Anatomy: rule [family] [source] [destination] [service|port|protocol] [log] [limit] [accept|reject|drop].
- accept vs reject vs drop: reject sends ICMP 'refused' (fast fail); drop is silent (stealth, slow timeouts).
- Rich rules beat opening SSH to the world — scope management ports to your IP ranges.

STEP 4/6

Runtime vs Permanent: Two Layers

Runtime is live-but-forgotten; permanent survives reboot.



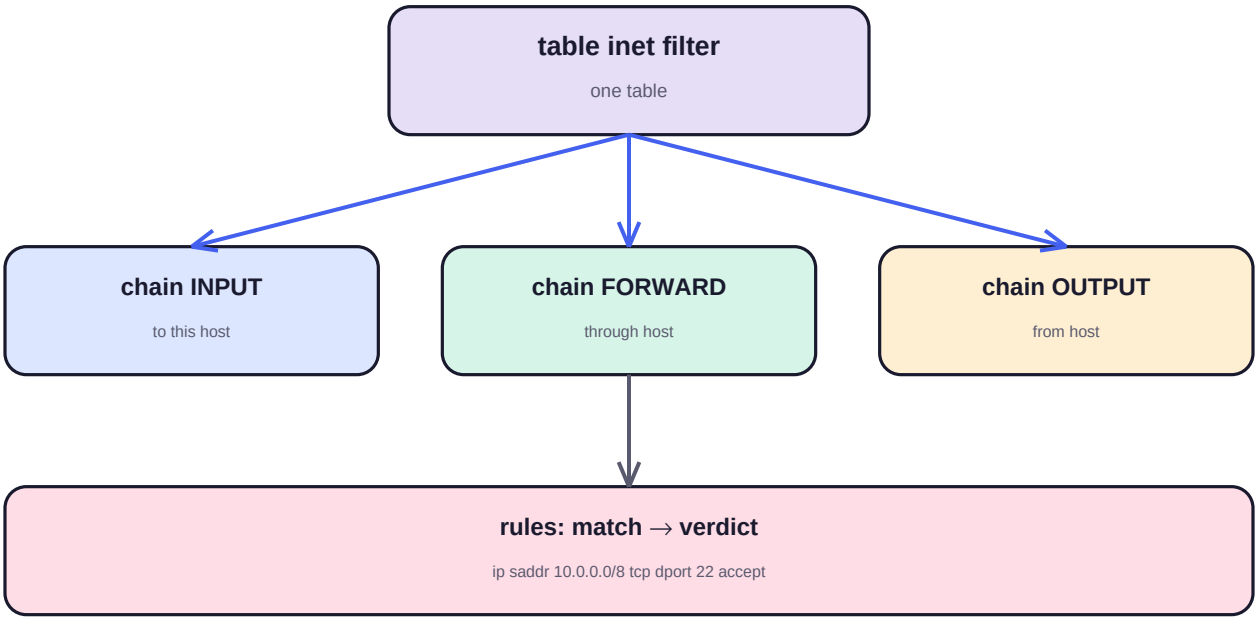

```
$ sudo firewall-cmd --add-service=http          # runtime only (test!)
$ sudo firewall-cmd --runtime-to-permanent      # keep what worked
$ sudo firewall-cmd --reload                    # apply permanent set
```

Workflow: test in runtime, verify the app works, THEN --runtime-to-permanent. A bad --permanent --reload over SSH can lock you out with no undo.

STEP 5/6

Under the Hood: nftables

firewalld is the friendly face; nftables is the engine.




```
$ sudo nft list ruleset          # see what firewalld actually built
$ sudo nft list chain inet filter INPUT
```

STEP 6/6

Firewall Triage Drill

Is it the firewall? Prove it in 60 seconds.

```
$ sudo firewall-cmd --state                    # running?
$ sudo firewall-cmd --list-all --zone=public   # what's open?
# test from the CLIENT:
$ nc -vz db.internal 5432     # refused = nothing listening OR reject
$ sudo tcpdump -i eth0 port 5432   # packets arrive but no reply = DROP
```

- Temporarily --set-target=ACCEPT on a test zone beats disabling firewalld (never do that).
- Remember BOTH firewalls: host firewalld AND cloud security groups / NACLs.
- Log dropped packets while debugging: rich rule with log prefix, then journalctl -k.

STEP ✓/✓

Key Takeaways

firewalld & nftables

- Zones assign trust per interface — the firewall follows the network.
- Prefer --add-service over raw ports; use rich rules for source-scoped exceptions.
- Runtime is for testing, permanent is for keeping — promote with --runtime-to-permanent.
- nftables tables → chains → rules is the engine; nft list ruleset shows the truth.
- Triage: state → list-all → client test → tcpdump; check cloud security groups too.

Keep exploring

Pair this guide with the matching video walkthroughs
and hands-on labs at interstitiumlabs.dev